

2 - Atoms, molecules and stoichiometry

07/12/24

Unified atomic mass is one twelfth of the mass of a carbon-12 (^{12}C) atom

- Relative atomic mass is the average mass of an atom compared to one twelfth of the mass of ^{12}C

$$A_r = \frac{\text{avg mass of one atom of an element}}{\frac{1}{12} \text{ mass of an atom of } ^{12}\text{C}}$$

- Relative isotopic mass is the average mass of an isotope compared to one twelfth of the mass of ^{12}C

- Relative molecular mass is the average mass of a molecule compared to one twelfth of the mass of ^{12}C

$$M_r = \frac{\text{avg mass of one molecule}}{\frac{1}{12} \text{ mass of an atom of } ^{12}\text{C}} \leftarrow \text{obtained by adding up all the } A_r$$

Relative formula mass is the average mass of the formula units compared to one twelfth of the mass of ^{12}C

! RMM is for molecular compounds while RFM is for compounds that aren't molecules

Mole

* the amount of substance that contains the same number of particles as there are in:

12 grams of carbon-12 (^{12}C)

this number of particles is called Avogadro's constant which is approximately 6.022×10^{23}

- Number of particles = moles of substance $\times$ Avogadro's constant

Q what is the number of moles in 35g of CuSO_4 ?

$$\text{moles} = \frac{\text{mass}}{M_r}$$

$$\text{mass} = 35$$

$$M_r = 63.5 + 32.1 + (16 \times 4) = 159.6$$

$$\text{moles} = \frac{35}{159.6}$$

$$= 0.219 \checkmark$$

Q how many atoms of tin are there in a 6g sample of tin?

$$\text{moles} = \frac{\text{mass}}{A_r}$$

$$\text{mass} = 6$$

$$A_r = 118.7$$

$$\text{moles} = \frac{6}{118.7}$$

$$= 0.0505$$

number of atoms

$$= 0.0505 \times [6.022 \times 10^{23}]$$

$$= 3.04 \times 10^{22} \checkmark$$

Q how many chloride ions are there in 25cm^3 of a solution of magnesium chloride with concentration 0.4 mol/dm^3 ?

* moles = conc. $\times$ volume *

$$25\text{cm}^3 \xrightarrow{(\div 1000)} 0.025\text{dm}^3$$

$$\text{moles} = 0.4 \times 0.025 = 0.01 \text{ mol}$$

$$0.01 \text{ mol of } \text{MgCl}_2$$

$$\text{Mg to Cl ratio} \rightarrow 1 : 2$$

$$\therefore 0.01 \times 2 = 0.02 \text{ mol of chloride ions}$$

$$\text{number of ions} = 0.02 \times [6.022 \times 10^{23}]$$

$$= 1.2 \times 10^{22} \checkmark$$

Ions to remember :

nitrate ion $\rightarrow \text{NO}_3^-$

carbonate ion $\rightarrow \text{CO}_3^{2-}$

sulfate ion $\rightarrow \text{SO}_4^{2-}$

hydroxide ion $\rightarrow \text{OH}^-$

phosphate ion $\rightarrow \text{PO}_4^{3-}$

ammonium ion $\rightarrow \text{NH}_4^+$

silver ion $\rightarrow \text{Ag}^+$

bicarbonate ion $\rightarrow \text{HCO}_3^-$

zinc ion $\rightarrow \text{Zn}^{2+}$

Constructing formulas

- deduce the formula of magnesium chloride

ions present: Mg^{2+} and Cl^-

for neutrality, we need two Cl^- ions for every Mg^{2+} ion $[2 \times -1] + [1 \times 2] = 0$

formula = MgCl_2

- write the formula of calcium sulfate

ions present: Ca^{2+} and SO_4^{2-}

for neutrality, we need one Ca^{2+} ion for one SO_4^{2-} ion

formula = CaSO_4

- name the compound Na_3PO_4

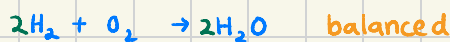
Na_3 PO_4
sodium + phosphate = sodium phosphate

- name the compound $(\text{NH}_4)_2\text{SO}_4$

$(\text{NH}_4)_2$ SO_4
ammonium + sulfate = ammonium sulfate

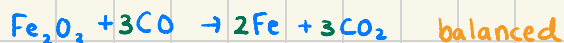
Constructing equations

- write a balanced equation for the reaction of hydrogen & oxygen $\rightarrow$ water

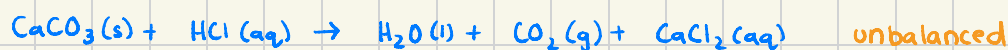


- write a balanced equation for the reaction of iron (III) oxide with carbon monoxide $\rightarrow$ iron & carbon dioxide

iron (III) oxide ions: Fe^{3+} & $\text{O}^{2-} \Rightarrow$ neutrality: two Fe^{3+} to three $\text{O}^{2-} = \underline{\underline{\text{Fe}_2\text{O}_3}}$

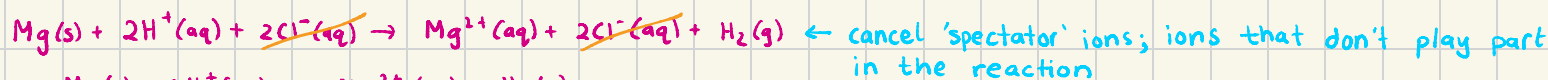
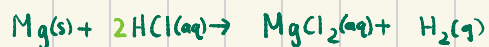


- solid calcium carbonate reacts with aqueous hydrochloric acid to form water, carbon dioxide & aqueous calcium chloride
write a balanced equation for this reaction.

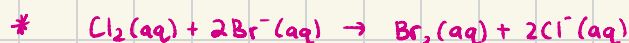
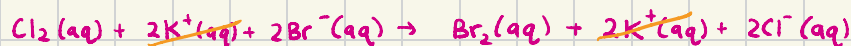


Note
state symbols:
(s) $\rightarrow$ solid
(l) $\rightarrow$ liquid
(g) $\rightarrow$ gas
(aq) $\rightarrow$ aqueous (solution)

write the ionic equation:



write the ionic equation:



Empirical formulae

* the simplest ratio of atoms of each element in a compound

Molecular formulae

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* the actual number of atoms of each element in a compound

calculate the empirical formula for a compound that contains 1.82g of K, 5.93g of I & 2.24g of O

$$\begin{array}{l} \text{K: } \frac{1.82}{39.1} = 0.0465 \\ \text{I: } \frac{5.93}{126.9} = 0.0467 \\ \text{O: } \frac{2.24}{16} = 0.14 \end{array} \left. \begin{array}{l} \frac{0.0465}{0.0465} = 1 \\ \frac{0.0467}{0.0465} = 1.00 \\ \frac{0.14}{0.0465} = 3.01 \end{array} \right\} \begin{array}{l} 1 : 1 : 3 \\ \text{KIO}_3 \checkmark \end{array}$$

÷ mass by atomic mass
÷ answers by the smallest answer

calculate the molecular formula for the compound with an empirical formula of $\text{C}_3\text{H}_6\text{O}$ & $M_r = 116$

Find the mass of the simple form

$$\text{C}_3\text{H}_6\text{O} : M_r = (12 \times 3) + (1 \times 6) + (16) = 58$$

$$\frac{116}{58} = 2$$

÷ the actual M_r by the simple form

$$\text{C}_3\text{H}_6\text{O} \times 2 = \text{C}_6\text{H}_{12}\text{O}_2 \checkmark$$

0.328g of a compound containing $\text{C}_3\text{H}_6\text{O}$ was burnt completely, producing 0.880g of CO_2 & 0.216g of H_2O . Calculate the empirical formula

$$\begin{array}{l} \text{CO}_2 : \text{moles} = \frac{0.880}{12 + (16 \times 2)} = 0.02 \text{ mol} \\ \text{C: } 0.02 \times 12 = 0.24 \text{ g} \\ \text{H}_2\text{O} : \text{moles} = \frac{0.216}{(1 \times 2) + 16} = 0.012 \text{ mol} \\ \text{H: } 0.012 \times 2 = 0.024 \text{ g} \end{array} \left. \begin{array}{l} \text{O: } 0.328 - 0.24 - 0.024 = 0.064 \text{ g} \\ \text{C: } \frac{0.24}{12} = 0.02 \\ \text{H: } \frac{0.024}{1} = 0.024 \\ \text{O: } \frac{0.064}{16} = 0.004 \end{array} \right\} \begin{array}{l} \frac{0.02}{0.004} = 5 \\ \frac{0.024}{0.004} = 6 \\ \frac{0.004}{0.004} = 1 \end{array}$$

5 : 6 : 1
 $\text{C}_5\text{H}_6\text{O} \checkmark$

Water of crystallisation

→ some compounds can form crystals which have water as part of their structure

→ this water is called the water of crystallisation

* a compound containing water of crystallisation is called a hydrated compound

e.g. hydrated copper (II) sulfate $[\text{CuSO}_4 \cdot 5\text{H}_2\text{O}]$

* a compound not containing water of crystallisation is called an anhydrous compound

e.g. anhydrous copper (II) sulfate $[\text{CuSO}_4]$

anhydrous → hydrated by adding water $\therefore \text{CuSO}_4 + 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

hydrated → anhydrous by heating $\therefore \text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 + 5\text{H}_2\text{O}$

Conversions & Formulas to memorize

for solids,
liquids &
gases

$$\left[\text{moles} = \frac{\frac{\text{g}}{\text{mol}}}{M_r} \right]$$

for
gases

$$\left[\text{moles} = \frac{\frac{\text{dm}^3}{24}}{\frac{\text{mol}}{\text{dm}^3}} \right]$$
$$\left[\frac{P}{\text{Pa}} = n \frac{R}{\text{mol}} \frac{T}{\text{K}} \right] \text{ \# unit 4}$$

for
solutions

$$\left[\text{conc.} = \frac{\frac{\text{mol/dm}^3}{\text{dm}^3}}{\text{dm}^3} \right]$$

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

$$C^\circ = K - 273$$

$$\text{cm}^3 \xrightleftharpoons[\times 1000]{\div 1000} \text{m}^3 \xrightleftharpoons[\times 1000]{\div 1000} \text{dm}^3$$